

Project-Q Workbook – 5

30-Day Quantum Computing Challenge

DAY – 05

Measurement & State Collapse

Why Measurement Is Special

Imagine you're watching a movie. Before pressing the **Pause** button, the movie is continuously changing. The moment you press **Pause**, one frame is frozen on the screen. Quantum measurement behaves in a somewhat similar way.

Before measurement:

A qubit can exist in a quantum state.

After measurement:

You observe exactly **one classical result**.

This process is called: **Quantum Measurement**

What Is Quantum Measurement?

Measurement is the process of reading information from a qubit. Unlike a classical bit, a qubit cannot simply be "looked at" without affecting its state.

When measured, the qubit produces one of the computational basis states:

$|0\rangle$ OR $|1\rangle$

No other outcome is possible for a single qubit measured in the computational (Z) basis.

Before vs After Measurement

Suppose a qubit is prepared as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Before measurement:

Superposition



Measure



$|0\rangle$ OR $|1\rangle$

After measurement, the superposition is no longer available.

Think About a Multiple-Choice Exam

Imagine a student has not yet submitted an online quiz.

The answers are still editable.

Draft Answers



Click "Submit"



Final Answers

After submission, the answers become fixed.

Similarly, before measurement, a qubit is in a quantum state. Measurement produces one definite outcome.

Does Measurement Always Give the Same Result?

No.

The result depends on the **probability amplitudes**.

Example:

$$|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Probabilities:

$$P(|0\rangle) = 50\%$$

$$P(|1\rangle) = 50\%$$

If you prepare this state many times and measure it repeatedly:

Run 1 $\rightarrow |0\rangle$

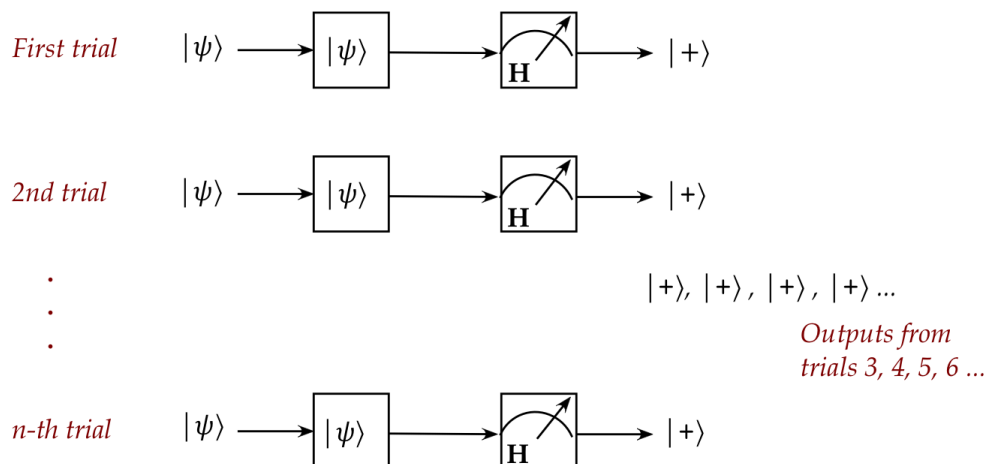
Run 2 $\rightarrow |1\rangle$

Run 3 $\rightarrow |0\rangle$

Run 4 $\rightarrow |1\rangle$

...

Over many measurements, the outcomes approach the predicted probabilities.



A Real Experiment

Imagine flipping a fair coin.

One flip:

Heads or Tails

You cannot predict a single flip.

But after **1,000 flips**:

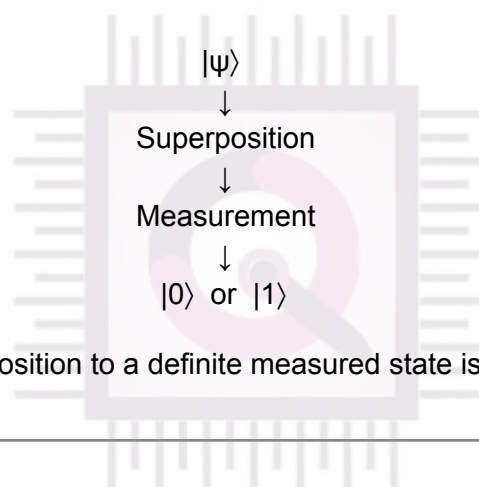
≈ 500 Heads

≈ 500 Tails

Similarly, a single quantum measurement is probabilistic, but repeated measurements reveal the probability distribution.

What Is State Collapse?

Before measurement:



The transition from a superposition to a definite measured state is called: **State Collapse**

Visualizing Collapse

Imagine a spinning arrow.

While spinning: Direction is changing continuously.

The moment you stop it: One fixed direction remains.

Similarly, a qubit evolves as a quantum state until measurement produces one observable outcome.

Important Rule

Suppose we measure:

$$(|0\rangle + |1\rangle)/\sqrt{2}$$

If the outcome is: $|0\rangle$

The qubit immediately becomes: $|0\rangle$

It is **no longer** in superposition.

Likewise, if the outcome is: $|1\rangle$

The new state becomes: $|1\rangle$

This is the meaning of state collapse.

Measuring Again

Suppose the first measurement gives: $|0\rangle$

Now measure again immediately.

Result: $|0\rangle$

Again.

And again.

$|0\rangle$

Why?

Because after the first measurement, the state has already collapsed to $|0\rangle$.

There is no remaining superposition to produce a different result.

Measurement Is Irreversible

Think about bursting a balloon.

Before:

Balloon



Pop!

After: Pieces of Rubber

You cannot return to the original balloon by simply looking at it.

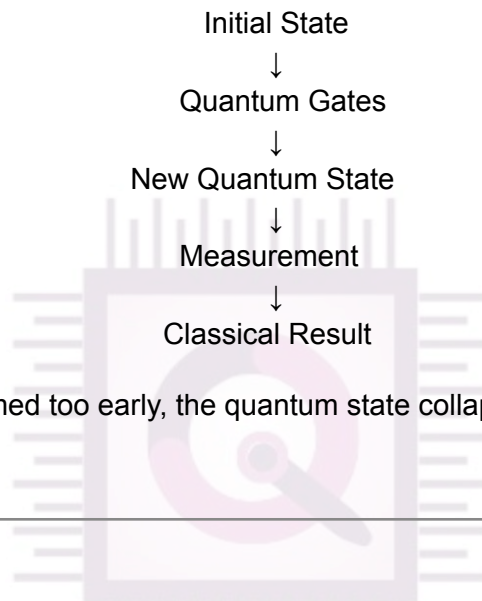
Similarly, Once a quantum state has been measured, the original superposition cannot be recovered from that measurement alone.

Why We Measure at the End

Quantum algorithms manipulate quantum states using quantum gates.

Only after all computations are complete do we measure.

The process looks like this:



If measurement is performed too early, the quantum state collapses before the computation is finished.

Measurement in Real Quantum Computers

On IBM Quantum hardware, measurement is represented by the **Measurement Gate**.

Every execution of a quantum circuit ends with measurement because classical computers can only process classical outputs.

Without measurement: No classical result can be observed.

Measurement Statistics

Suppose we execute the same quantum circuit **1,000 times**.

Possible result:

	Outcome	Counts
0	506	
1	494	

Instead of looking at a single measurement, quantum programmers analyze many measurements.

These repeated executions are called **shots**.

More shots generally provide a better estimate of the underlying probabilities.

Common Beginner Misconceptions

✗ Misconception 1

Measurement reveals the hidden value of the qubit.

✓ Reality

The measurement outcome is governed by the quantum state's probability amplitudes.

✗ Misconception 2

Measurement destroys the qubit.

✓ Reality

Measurement changes the state into one of the measurement basis states.

The qubit still exists; it is now in the measured state.

✗ Misconception 3

Measuring twice gives different answers.

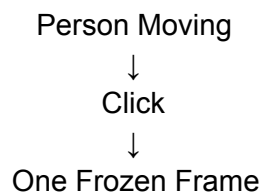
✓ Reality

If nothing changes between measurements, repeated measurements produce the same result because the state has already collapsed.

Real-World Analogy

Imagine taking a photograph.

Before the picture:



The photograph captures one moment.

Similarly, Measurement captures one outcome from the quantum state according to its probability distribution.

Quick Comparison

Concept	Meaning
Measurement	Reading information from a qubit
Superposition	Multiple basis states combined into one quantum state
Probability	Likelihood of each measurement outcome
State Collapse	Transition from a quantum state to one definite basis state after measurement
Shots	Repeating the same circuit many times to estimate probabilities
Measurement Basis	The set of states used to perform a measurement (commonly the computational Z basis)

End-of-Day Summary

What You Learned Today

- ✓ What quantum measurement is
- ✓ Why measurement differs from classical reading

- ✓ Measurement outcomes
 - ✓ State collapse
 - ✓ Why repeated measurements produce statistics
 - ✓ Why measuring twice gives the same result
 - ✓ Why quantum algorithms measure at the end
 - ✓ Measurement in IBM Quantum circuits
 - ✓ Understanding "shots"
-

Top Verified Learning Resources

1. IBM Quantum Learning (Highest Priority)

Quantum States and Measurement

[IBM Quantum Learning Platform](#)

Study the sections on:

- Quantum States
- Measurement
- Superposition
- Quantum Circuits

This is IBM's official educational platform and closely matches how measurement is implemented on real quantum hardware.

2. IBM Quantum – Utility Scale Quantum Computing

[Bits, Gates and Circuits Course](#)

Focus on:

- Measurement gates
 - Circuit execution
 - Shots
 - Classical outputs
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3. Microsoft Azure Quantum Documentation

[Azure Quantum Learning Documentation](#)

Recommended topics:

- Measurement
 - Quantum states
 - Probability amplitudes
 - Quantum circuits
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4. Qiskit Documentation

[Qiskit Documentation – Measurement Guide](#)

Learn how measurement is represented in quantum circuits and how results are collected using shots.

Video Resources

IBM Quantum YouTube

[IBM Quantum YouTube Channel](#)

Recommended videos:

- *Quantum Measurement Explained*
 - *Understanding Qubits and Measurement*
-

Qiskit YouTube

[Qiskit YouTube Channel](#)

Look for beginner videos covering:

- Quantum Measurement
- State Collapse
- Measurement in Quantum Circuits